

MODEL ANSWERS

Inha Univeristy in Tashkent
Quiz -1

MM[10]

Course Name: Computer Security

Section:

Student ID : U _____

NAME _____

Q1. Ciphertext is calculated as $C = (5P + 8) \bmod 26$.

Encrypt plaintext numeric values: 3, 0, 19, 0. Convert final result to letters.

Answer:

$$15 + 8 = 23 \bmod 26 = X$$

$$8 = 8 \bmod 26 = I$$

$$9 \cdot 5 + 8 = 103 \bmod 26 = 25 = Z$$

$$8 = 8 \bmod 26 = I$$

X I Z I

Q2. In DES S-box (S1) Given Input = 101011, Find the S1 output (4 bits).

Answer:

$$\text{row} = 11$$

$$\text{col} = 0101 = 5$$

$$09 = 1001$$

Q3. Using Fermat's Little theorem find $13^{2010} \bmod 71$.

Answer:

$$13^{2010} \bmod 71 \quad 13 \perp 71$$

$$13^{70 \times 28} \cdot 13^{50} \bmod 71 = 13^{50} \bmod 71 =$$

Q4. Find $\Phi(455)$

Answer:

$$288$$

$$(5 \times 13 \times 7)$$

5. Using Playfair key **SECURITY**, encrypt digraphs:

$$1. SE \rightarrow EC$$

$$2. CU \rightarrow UR$$

Answer:

S	E	C	U	R
I/J	T	Y	A	B
D	F	G	H	K
L	M	N	O	P
Q	V	W	X	Z

MODEL ANSWERS

Inha University in Tashkent Quiz -1

Course Name: Computer Security

MM[10]

Section:

Student ID: U _____

NAME _____

Q1. Ciphertext is calculated as $C = (7P + 8) \bmod 26$.

Encrypt plaintext numeric values: 3, 0, 19, 0. Convert final result to letters.

Answer: $21 + 8 \bmod 26 = 29 \bmod 26 = 3 \rightarrow D$
 $8 \bmod 26 = 8 \rightarrow I$
 $14 + 8 \bmod 26 = 22 \bmod 26 = 22 \rightarrow L$
 $8 \bmod 26 = 8 \rightarrow I$

Q2. In DES S-box (S2) Given Input = 011110, Find the S2 output (4 bits).

Answer: Row 00 - 0
Col 1111 - 15 $\Rightarrow 10 = 1010$

Q3. Find $\Phi(1200)$

Answer:

$$\phi(1200) = \phi(2^4 \times 3 \times 5^2) = (2^4 - 2^3)(5^2 - 5)(2) = 8 \times 20 \times 2 = 320$$

Q4. Using Fermat's Little theorem find $(2^{444} + 3^{333}) \bmod 5$.

Answer:

$2^4 \equiv 1 \pmod{5}$ (1 times) $3^4 \equiv 3 \pmod{5}$ (83 times)
 $1 + 3 = 4$

Q5. Encrypt NETWORK using shift $k = 17$.

N	E	T	W	O	R	K
13	4	19	22	14	17	10

+ 17

30	21	36	39	31	34	27
----	----	----	----	----	----	----

) mod 26

4	21	10	13	5	8	1
E	V	K	N	F	I	B

MODEL ANSWERS

Inha Univeristy in Tashkent Quiz -1

Course Name: Computer Security

MM[10]

Section: Student ID: U _____

NAME _____

Q1. Ciphertext is calculated as $C = (3P + 7) \bmod 26$.

Encrypt plaintext numeric values: 3, 0, 19, 0. Convert final result to letters.

Answer: $3 + 7 = 10 \bmod 26 = 10 \rightarrow Q$
 $0 = 0 \bmod 26 = 0 \rightarrow H$
 $19 = 19 \bmod 26 = 19 \rightarrow M$
 $0 = 0 \bmod 26 = 0 \rightarrow H$
Q H M H

Q2. In DES S-box (S1) Given Input = 011110, Find the S1 output (4 bits).

Answer: $\begin{matrix} 00 \\ 1111 \end{matrix} \Bigg] 07 \rightarrow 0111$

Q3. Find $\Phi(2008)$

Answer: $\phi(2^3) \phi(251)$
 $4 \cdot 250 = 1000$
1000

Q4. Solve the congruence

$$x^{103} = 4 \bmod 11. \text{ Find } x.$$

$$x \in \mathbb{Z}_{11}$$

Answer: $x^{10} \bmod 11 = 1$

$$(x^{10})^{10} x^3 \bmod 11 = 4 \bmod 11$$

$$\text{per} = 5$$

$$x = 5$$

Q5. A ciphertext **GUR DHVPX OEBJA SBK** is encrypted using a shift cipher. Find the plaintext if key $k = 13$.

G	U	R	D	H	V	P	X	O	E	B	J	A	S	B	K
6	20	17	3	7	21	15	23	14	4	1	9	0	18	1	10

-13

- 7 7 4 -10 -6 8 2 10 1 -9 -12 -4 -13 5 -12 -3

19 7 4 16 20 8 2 10 1 17 14 22 13 5 14 23

T H E Q U I C K B R O W N F O X

T H E Q U I C K B R O W N F O X

MODEL ANSWERS

Inha University in Tashkent Quiz -1

Course Name: Computer Security

Section:

Student ID : U _____

NAME _____

MM[10]

Q1. Ciphertext is calculated as $C = (4P + 9) \bmod 26$.

Encrypt plaintext numeric values: 3, 0, 19, 0. Convert final result to letters.

Answer: $E(3) = (4 \cdot 3 + 9) \bmod 26 = 21 \bmod 26 = 21 - V$
 $C(0) = 9 \bmod 26 = 9 - J$
 $C(19) = (4 \cdot 19 + 9) \bmod 26 = (76 + 9) \bmod 26 = 7 - H$ J

Q2. In DES S-box (S1) Given Input = 111100, Find the S1 output (4 bits).

Answer: Row - 2
Col - 14 $\rightarrow 5 - 0101$

Q3. Using Fermat's Little theorem find $(3^{444} + 4^{333}) \bmod 5$.

Answer: $3^{444} \bmod 5 = 3^4 - 11 \text{ times} = 0$
 $4^4 \bmod 5 = 4$ $4 + 1 \bmod 5 = 0$

Q4. Find $\Phi(616)$

Answer: $\phi(616) = \phi(2^3) \phi(7) \phi(11)$
 $= 4 \times 6 \times 10$
 $= 240$

Q5. Encrypt SECURITY using shift key $k = 11$, show intermediate numeric conversion.

Answer:

S	E	C	U	R	I	T	Y
18	4	2	20	17	8	19	24
29	15	13	31	28	19	30	35
3	15	13	5	2	19	4	9
D	P	N	F	C	T	E	J